

2-adic even blow-up Tate normal form $y^2 + xy = x^3 + a_2x^2 + a_4x + a_6$ — three-open gluing v33

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Abstract

This working v33 manuscript records a three-open **OpenCover** of the *quotient* Rees scheme $\text{Proj}(\text{ReesMod2})$ by $D_+(2t)$, $D_+(\overline{X}t)$, and $D_+(\overline{Y}t)$, with affine restriction squares for every ordered pair over abstract product-open pullbacks. **CompatChart2t** (191 lines) identifies the principal divided-chart kernel (F) and proves **chart_eq_quotient_2t**; **CompatChartXt** and **CompatChartYt** (105 lines each) prove $J_X = (G_X : X^\infty)$ and $J_Y = (G_Y : Y^\infty)$ and their quotient-induced chart identifications. Six directed polynomial-to-abstract restrictions satisfy $\text{res}_{\text{poly}} = \text{res}_{\text{actual}} \circ \psi$. **SpecialFibreGluing** (167 lines) checks six composed **Spec.map** compatibilities and the abstract triple cocycle, giving

$$\text{Proj}(\text{ReesMod2}) \cong \text{Glue}(S_{2t}/(2), S_{Xt}/(2), S_{Yt}/(2)).$$

Product overlaps are abstract pullbacks, not polynomial presentations. For $y^2 + xy = x^3 + 8$, the reduced $2t$ chart is $V^2 + UV = 0$; the point $U = V = 0$ lies only in $D_+(\overline{2}t)$ (an illustration, not a separately named Lean theorem). **TateEvenBranchGenericFibre** (59 lines) packages the *actual* blow-up comparison $\text{Bl}_I[1/2] \cong \text{Spec}(R_Z[1/2])$ with $I[1/2] = \mathbb{T}$; the existing 2,249-line chart file is untouched. The seven targeted **lake env lean** direct checks previously passed with printed axioms only **propext**, **Classical.choice**, and **Quot.sound**; the scoped **sorry** scan and **git diff --check** are clean. A standalone direct check of the large special-fibre file timed out without a diagnostic and a full **lake build BealGeneral** remains unverified. We do *not* claim a global isomorphism from the actual special-fibre pullback to the quotient Rees Proj, or a degree-one quotient formula or nonvanishing theorem for $\overline{2}t$. This is not a general Beal proof. The intended release and Zenodo source is **beal-conjecture**, not its level-26 work mirror; its active GitHub-side Zenodo **release** webhook does not yet establish delivery, a deposit, or a DOI. A DOI may be minted only after an authorized GitHub Release v33 and subsequent Zenodo verification.

1 Scope and charts

Write $R_Z = \mathbb{Z}_2[X, Y]/(f)$ for the translated even-branch surface under the stated nodal and coefficient-divisibility hypotheses, $I = (2, X, Y)$, and $\text{ReesMod2} = \text{Rees}(I)/(2)$, where (2) is generated by the degree-zero base scalar, *not* by $2t$. The integral divided chart has $f(2U, 2V) = 4F$; its principal relation is (F) . The Xt and Yt charts use saturated graph ideals $(G_X : X^\infty)$ and $(G_Y : Y^\infty)$ rather than the unsaturated graph equations. Quotient-induced chart maps connect these integral calculations to the three reduced chart spectra. These assertions concern the charts under their hypotheses, not an unconditional formula for every Weierstrass model.

2 The checked gluing boundary

The three reduced generator opens cover $\text{Proj}(\text{ReesMod2})$. Ordered affine restriction squares live over product opens; the six distinct directed pairs obey the polynomial-to-abstract restriction identities. The polynomial spectra are literal cover objects, but their *overlaps* are abstract product pullbacks. Six Spec.map squares and a triple cocycle yield the global isomorphism displayed in the abstract. No polynomial ring presentation of a product overlap is supplied or inferred.

The illustrative $y^2 + xy = x^3 + 8$ specialization has $f(2U, 2V) = 4(V^2 + UV - 2U^3 - 2)$, so its reduced divided-chart equation is $V^2 + UV = 0$. The point $(U, V) = (0, 0)$ gives direction $[2t : Xt : Yt] = [1 : 0 : 0]$ and belongs to the $2t$ open alone; this is an example, not a global classification theorem.

3 Generic fibre and unresolved special-fibre comparison

After inverting 2, the centre I becomes the unit ideal. The packaged generic-fibre isomorphism concerns the *actual* blow-up and the localized translated surface over $D(2)$; it does not identify an integral chart with its generic fibre. In contrast, the checked special-fibre gluing concerns the *quotient Rees Proj*. Chartwise base-change comparisons do not yet give an isomorphism of that Proj with the actual scheme-theoretic special-fibre pullback. Nor is $(\text{ReesMod2})_1 \cong (I/2I)t$ or $\overline{2t} \neq 0$ certified here. These missing steps are not used as premises of the quotient-*Proj* gluing.

4 Verification and deposit route

The seven direct Lean checks and their axiom reports are recorded in `Beal/RELEASE_NOTES_v33.md` and `Beal/AUDIT_v33_HONESTY.md`. They are not a verified full-target build: the standalone large special-fibre source check timed out, although the passing modules imported its compiled declarations. The existing `TateEvenBranch.lean` (2,249 lines) passed its separate direct check and was not refactored.

`beal-conjecture` is the whole-project main audit and intended GitHub-release source. The `beal-level-26-foundations` work repository is the later mirror for the level-26 scaffold; keep that scaffold separate rather than present it as an inhabited global Beal result. An authenticated GitHub-side check found one active Zenodo hook subscribed to `release` events for the main repository. Neither a tag nor a release nor a v33 DOI exists yet. Only after explicit release approval should the resulting Zenodo record, archive, metadata, and DOI be checked; no mirror push is authorized by this manuscript.